Biomagnification of DDT

When non-degradable substances enter the food chain, they do not get metabolized or broken down or expelled and instead get transferred up the tropic levels of the food chain. During this process, they show an increase in concentration which is referred to as **biomagnification**. This results in increased toxicity and may even be lethal. This phenomenon is well established for mercury and DDT. **Fig 12.4** schematically shows biomagnification of DDT in an aquatic food chain where the concentration of DDT is enhanced at successive trophic levels.

12.7 Eutrophication

When run-off from land containing nutrients reaches water bodies like lakes, it results in dense growth of plant life. This phenomenon is called **Eutrophication**. Natural aging of lakes also leads to nutrient enrichment of its water. In a lake, the water is cold and clear (**oligotrophic stage**), supporting little life. With time, streams draining into the lake introduce nutrients such as nitrates and phosphates, which encourage the growth of aquatic organisms. Aquatic plants and animal life grow rapidly, and organic remains begin to be deposited on the lake bottom (**mesotrophic stage**) **(Fig. 12.5).**

Pollutants from anthropogenic activities like effluents from the industries and homes can radically accelerate the aging process. This phenomenon is known as **Cultural** or **Accelerated Eutrophication.**

Nutrients stimulate the growth of algae, water hyacinth and can cause clogging of canals, rivers and lakes as well as, displacing native plants. It causes unsightly foam and unpleasant odours, and deprives the water of dissolved oxygen.

12.7.1 Integrated Wastewater Management

Wastewater Treatment

Wastewater or sewage originates from domestic waste waters, industrial wastes and animal wastes. Realizing the importance of clean potable water, the Government passed the Water (Prevention and Control of Pollution) Act in 1974, which made it mandatory to treat wastewater in treatment plants. The main objective of a wastewater treatment process is to reduce organic and inorganic components in wastewater to a level that it no longer supports microbial growth and to eliminate other potentially toxic materials. Microorganisms mainly bacteria and some protozoa play an essential part in the treatment of sewage to make it harmless. Sewage contains pathogenic bacteria. These bacteria must be destroyed in order to prevent the spread of diseases. Sewage treatment is usually performed in the following three stages **(Fig. 12.6)**.

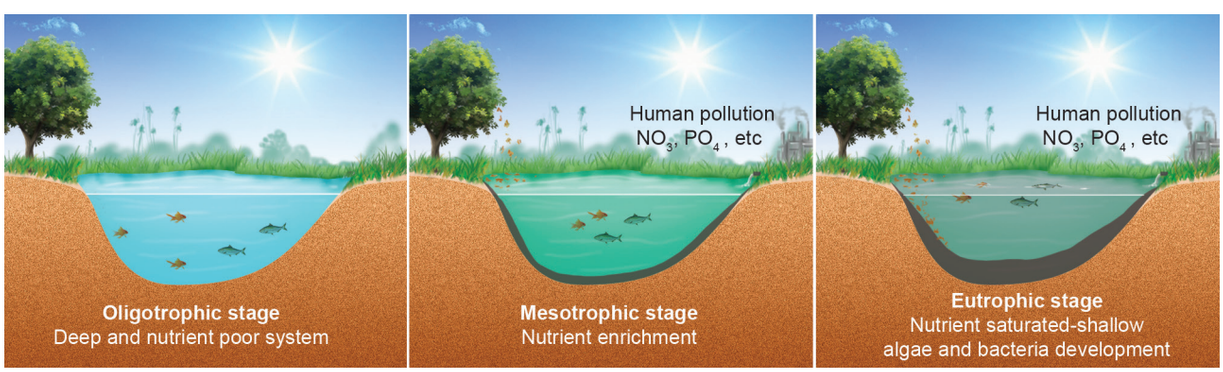


Fig. 12.5 Stages of Eutrophication

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